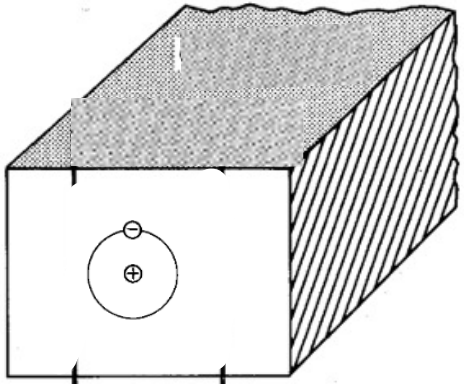
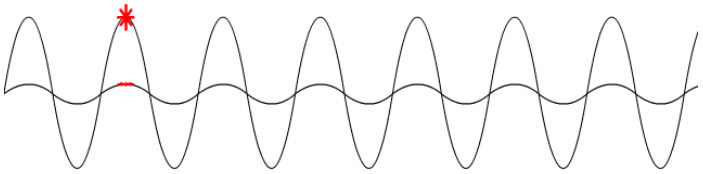
EM Waves-matter Interaction

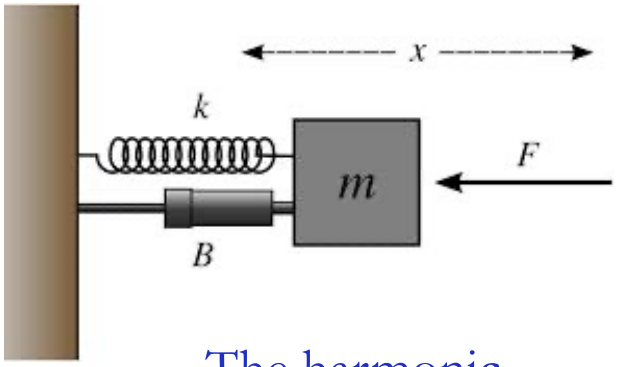
The following subjects will be covered

- EM wave-material Interaction.
- Complex Dielectric Constant.

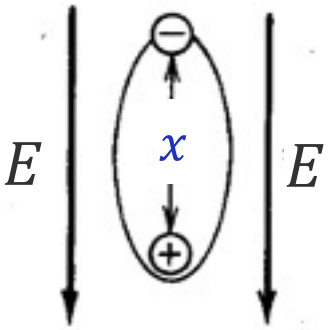
EM Wave-matter Interaction



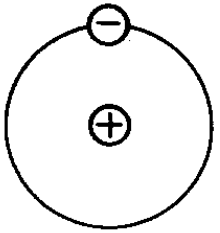




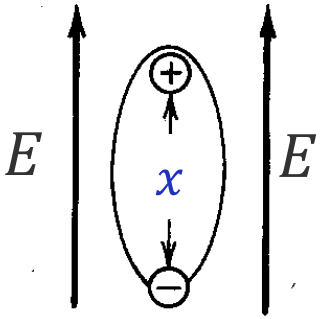
The harmonic oscillator



Atom polarized

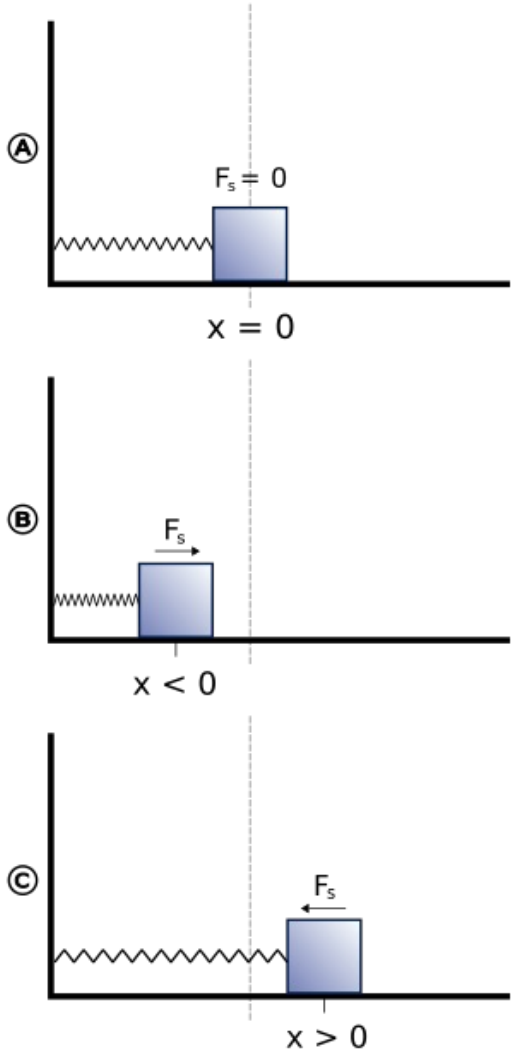


No Wave



Atom polarized

EM Wave-matter Interaction



$$m\frac{\partial^2\vec{x}}{\partial t^2} + m\Gamma\frac{\partial\vec{x}}{\partial t} + m\omega_0^2\vec{x} = -q\vec{E}$$

- $m\frac{\partial^2\vec{x}}{\partial t^2}$: acceleration force; $m \Rightarrow m_e$ mass of an electron
- $m\Gamma\frac{\partial\vec{x}}{\partial t}$: frictional force; Γ damping rate (loss/sec)
- $m\omega_0^2\vec{x}$: restoring force; $\omega_0 = \sqrt{\frac{K}{m}}$ natural frequency
- $-q\vec{E}$: electric force

Fourier transform

$$m(-j\omega)^2\vec{x}(\omega) + m\Gamma(-j\omega)\vec{x}(\omega) + m\omega_0^2\vec{x}(\omega) = -q\vec{E}(\omega)$$

Simplify

$$\left(-m\omega^2 - j\omega m\Gamma + m\omega_0^2\right)\vec{x}(\omega) = -q\vec{E}(\omega)$$

EM Wave-matter Interaction

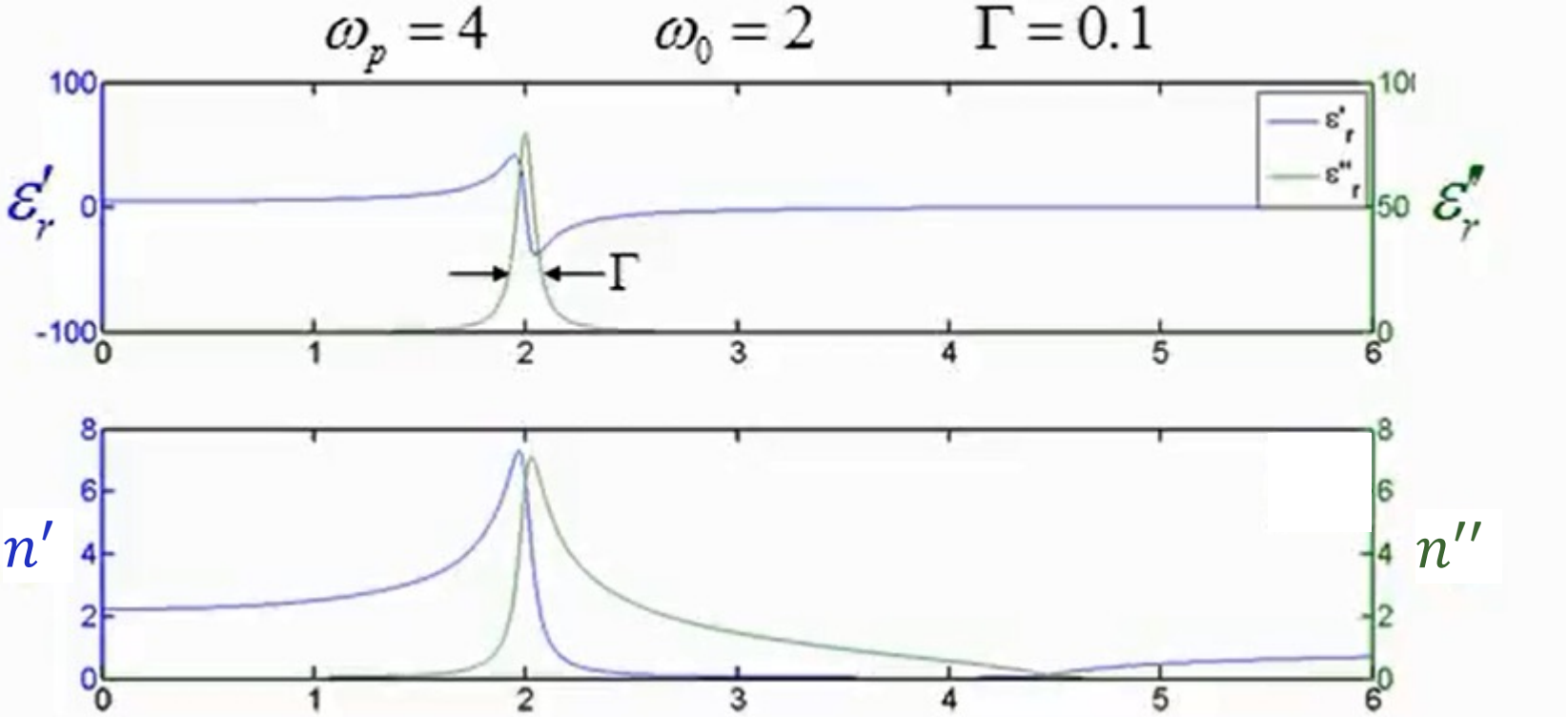
$$\varepsilon_r(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - j\omega\Gamma} \qquad \omega_p^2 = \frac{Nq^2}{\varepsilon_0 m_e}$$

$$\varepsilon_r(\omega) = \varepsilon_r'(\omega) - j\varepsilon_r''(\omega)$$

$$\varepsilon_r'(\omega) = 1 + \omega_p^2 \frac{\omega_0^2 - \omega^2}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

$$\varepsilon_r''(\omega) = \omega_p^2 \frac{\omega\Gamma}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

EM Wave-matter Interaction

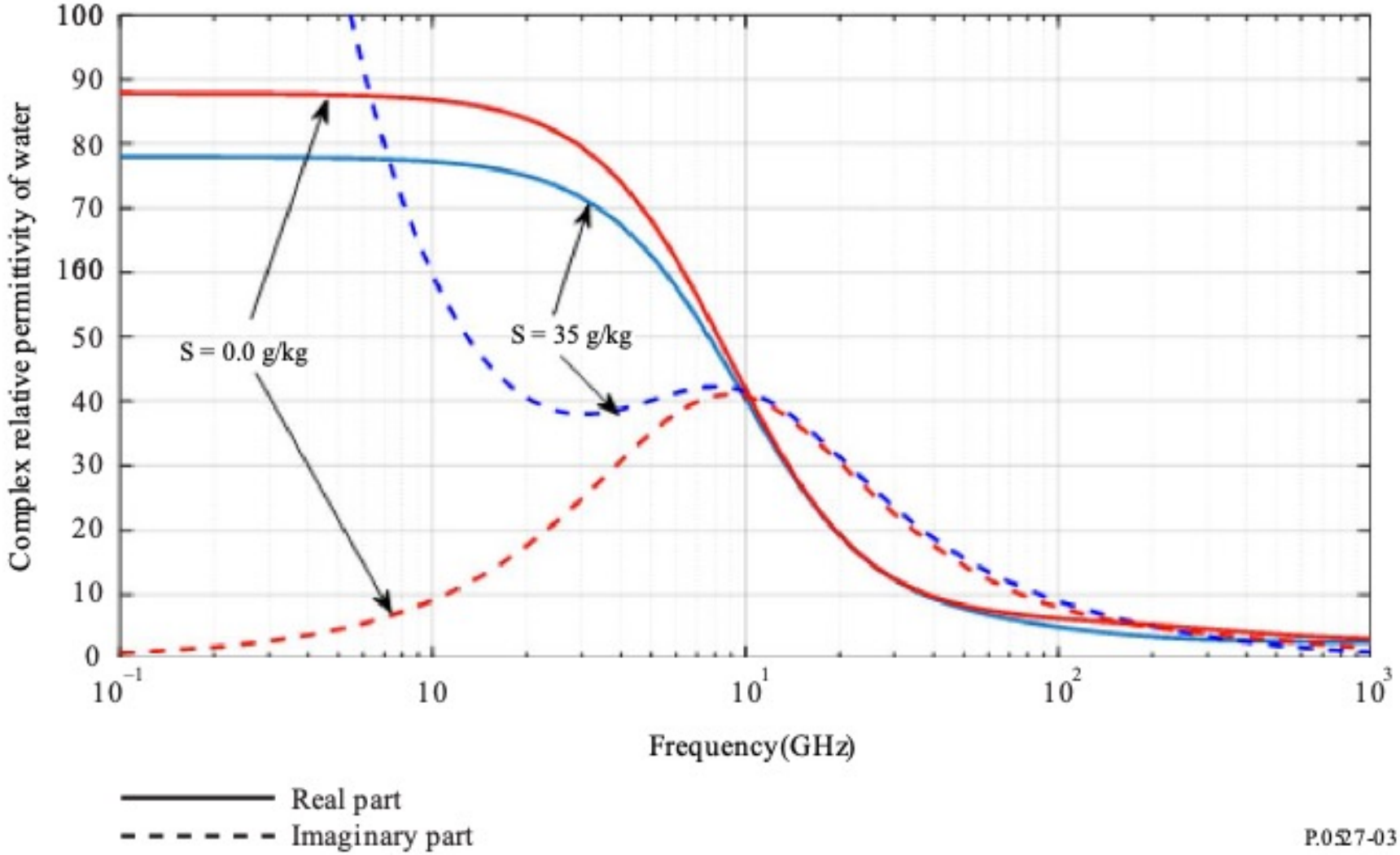


Loss is very high near resonance

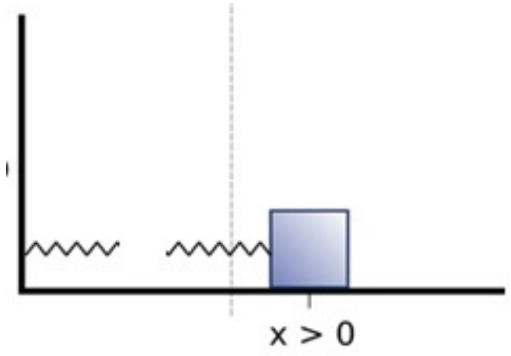
Damping rate determines width of resonance

Dielectric constant has a DC offset below resonance

Sea Water Dielectric Function



EM Wave-matter Interaction: Metals



$$m\frac{\partial^2 \vec{x}}{\partial t^2} + m\Gamma\frac{\partial \vec{x}}{\partial t} + m\omega_0^2\vec{x} = -q\vec{E}$$

acceleration force

$m \Rightarrow m_e$ mass of an electron

frictional force

Γ damping rate (loss/sec)

restoring force

$\omega_0 = \sqrt{\frac{K}{m}}$ natural frequency

electric force

$$\varepsilon_r(\omega) = 1 - \frac{\omega_p^2}{\omega^2 - j\omega\Gamma}$$

$$\varepsilon_r = \left(1 - \frac{\omega_p^2\tau^2}{1+\omega^2\tau^2}\right) + j\left(\frac{\omega_p^2\tau/\omega}{1+\omega^2\tau^2}\right)$$

Drude Model

Complex ε_r

We can combine the conductivity σ in the definition of the dielectric constant ε_r . In this case, a complex dielectric constant.

The time-harmonic Maxwell's equation is $$\vec{\nabla} \times \vec{H} = \sigma \vec{E} + j\omega\varepsilon \vec{E}$$

Combine terms $$\vec{\nabla} \times \vec{H} = (\sigma + j\omega\varepsilon)\vec{E} = j\omega\,\varepsilon\left(1 - \frac{j\sigma}{\omega\varepsilon}\right)\vec{E}$$

ε_{rc}

Compare to the lossless equation $$\varepsilon_{rc} = \varepsilon\left(1 - \frac{j\sigma}{\omega\varepsilon}\right) = \varepsilon_o\varepsilon_r\left(1 - \frac{j\sigma}{\omega\varepsilon_o\varepsilon_r}\right)$$

$$\varepsilon_c = \varepsilon_o\left(\varepsilon_r - j\frac{\sigma}{\omega\varepsilon_o}\right)$$

ε_{rc}

$$\varepsilon_{rc} = \varepsilon_r' - j\varepsilon_r''$$

Complex dielectric constant

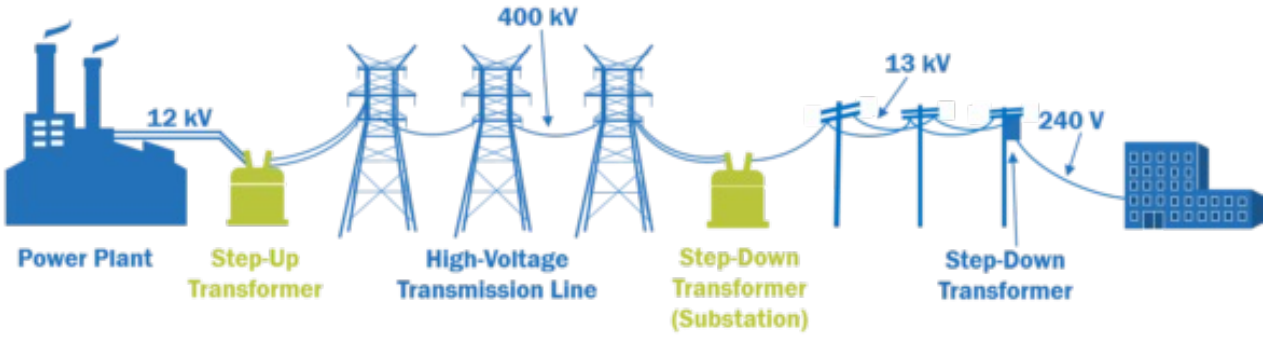
Transmission Lines

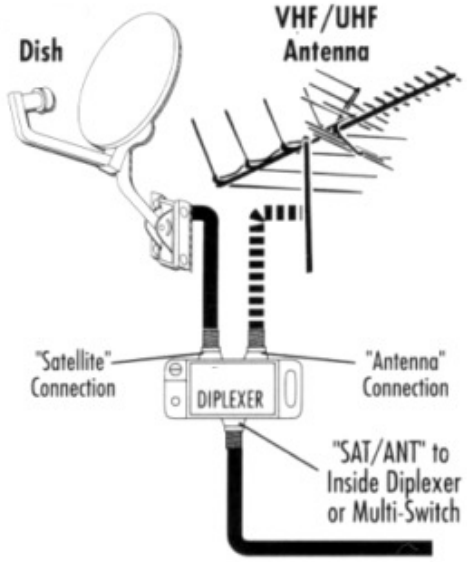
The following subjects will be covered

- Introduction.
- Transmission lines types
- The TL circuit model and equations.
- Wave propagation (guiding) on a TL.
- Analysis and examples.

Transmission Lines: Introduction

Cables/conductors that guide EM waves usually from source to load over long distances.



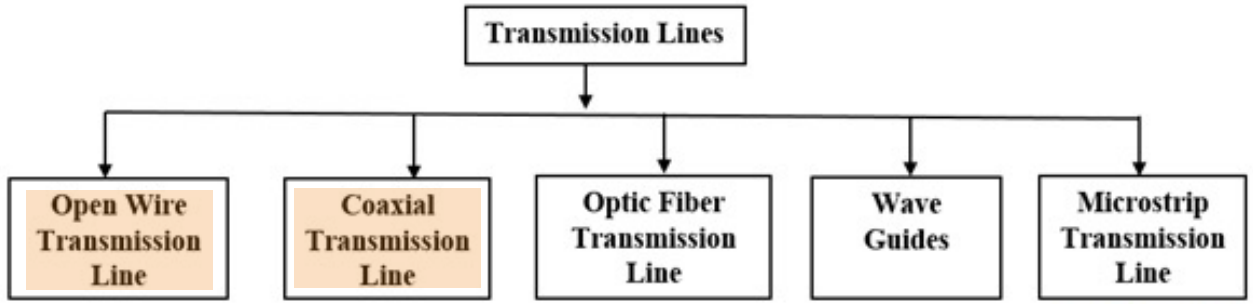


Electromagnetic analysis (waves) vs. Circuit analysis (voltage and current).
Which one?

1) Time-varying sources (charges, currents, etc.)
2) Frequency of operation.
3) Distance/wavelength ratio.

Transmission Lines: Introduction

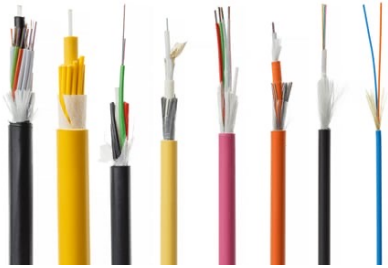
Any guiding device can be considered as a transmission line. However, TLs are classified according to domain of operation and principle of operation.

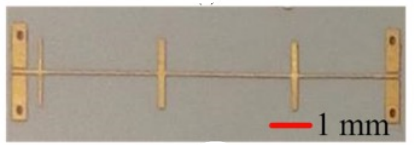


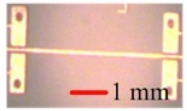


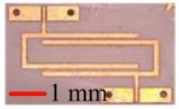






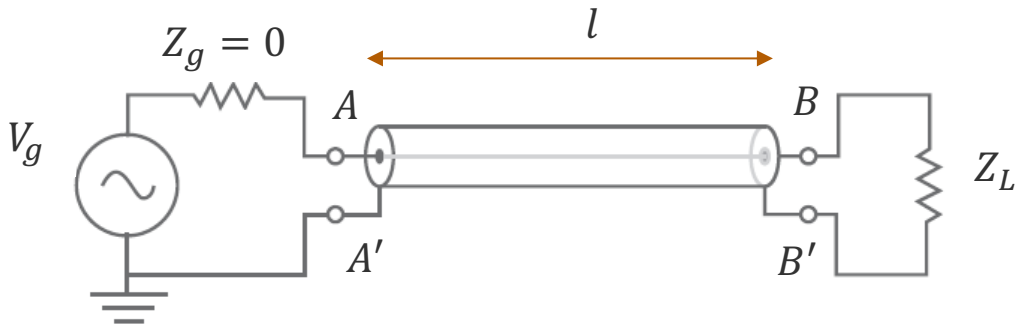






Transmission Lines: Introduction

Let's have a closer look at wave propagation on a transmission line.



$$V_g(t) = V_o \cos(\omega t) = V_{AA'}$$

$$V_{BB'}(t) = V_{AA'}(t - l/v)$$

$$V_{BB'}(t) = V_o \cos[\omega(t - l/v)]$$

$$= V_o \cos(\omega t - \omega l/v)$$

Phase shift

Take $t = 0$, for simplicity

For an air line $f = 1\ kHz, \quad l = 5\ cm.$

$V_{AA'} = V_o \qquad V_{BB'} = 0.999999\ V_o$

For $l = 20\ km.$

$V_{BB'} = 0.91\ V_o$

It's all about the phase shift

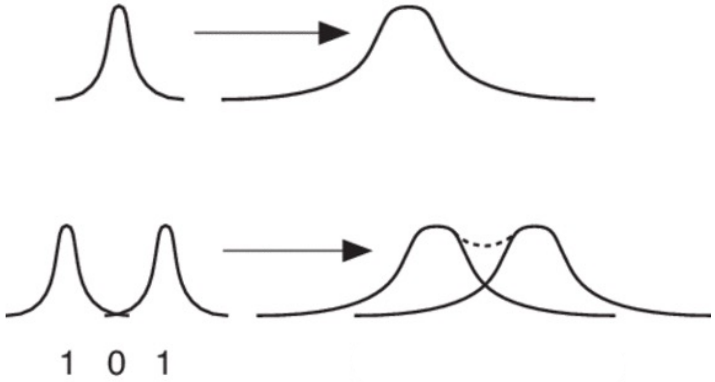
$$\frac{\omega l}{v} = 2\pi \frac{l}{\lambda}$$

Transmission Lines: Introduction

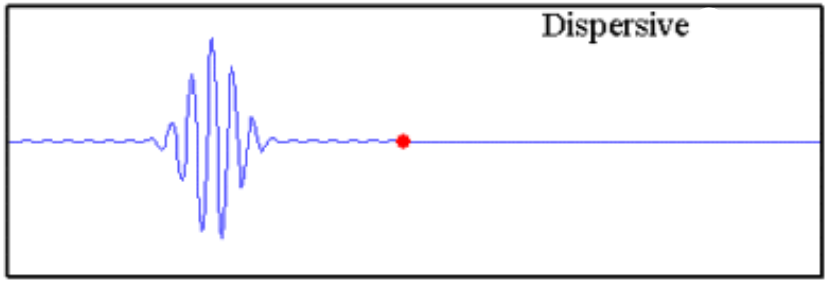
Dispersion: phase shift depends on wave frequency.

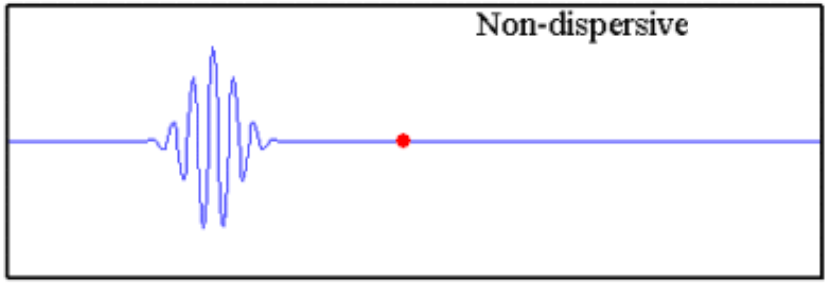


Non-dispersive line



Dispersive line

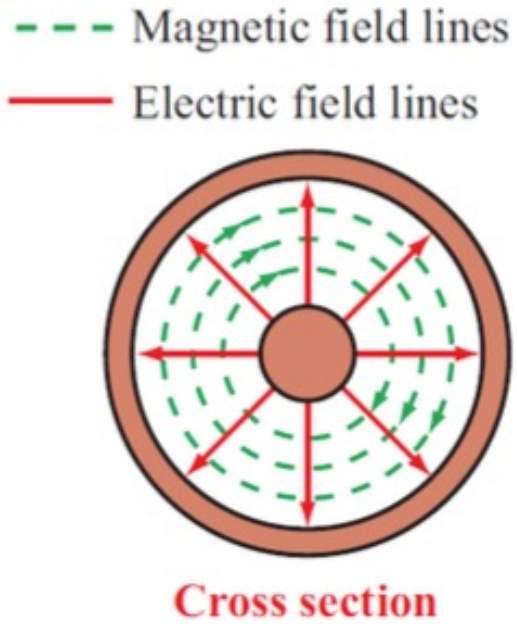




Transmission lines are designed to be lossless (ohmic and radiative) and nondispersive.

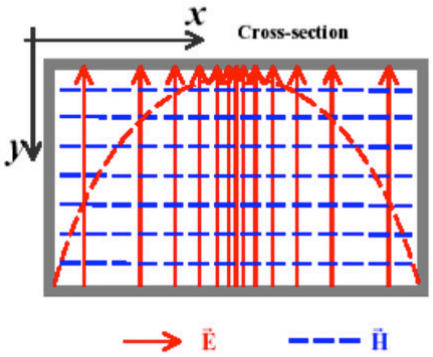
TEM Transmission Lines

Transmission lines are also classified based on the fields shape (modes) in the direction of propagation, as TEM and non-TEM.

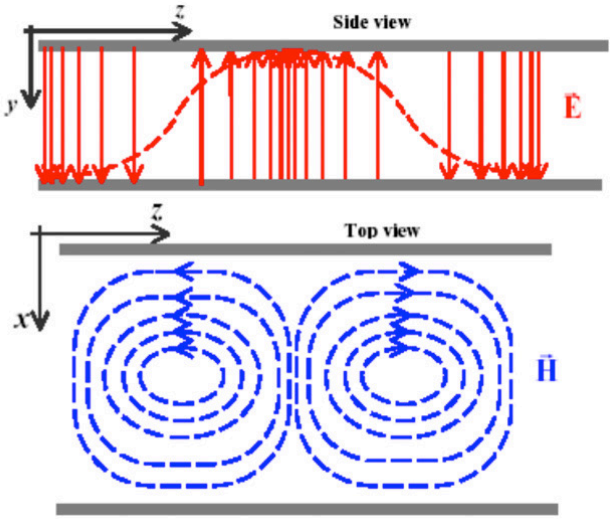




TEM



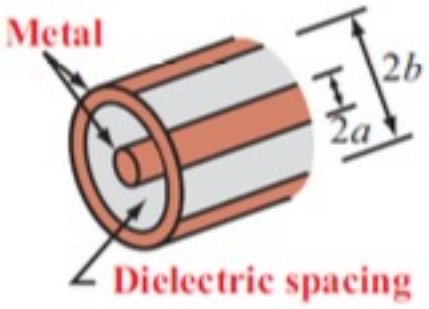




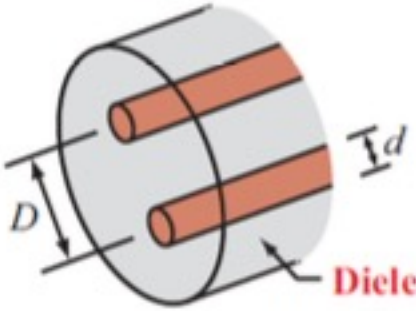
Non-TEM

TEM Transmission Lines

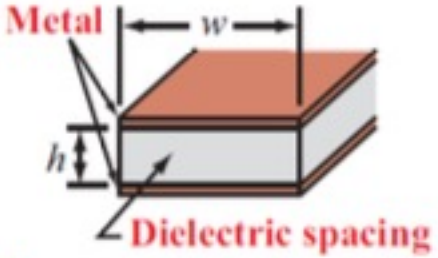
Examples of TEM transmission lines.



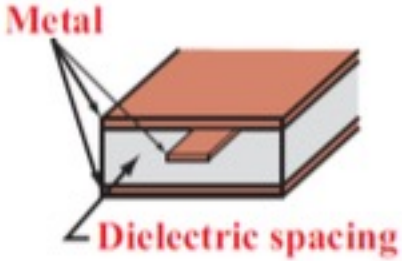
(a) Coaxial line



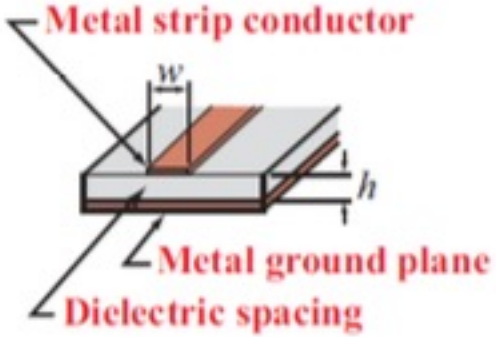
(b) Two-wire line



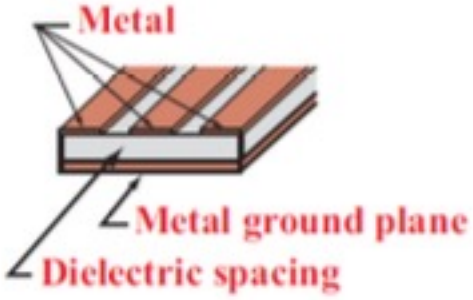
(c) Parallel-plate line



(d) Strip line



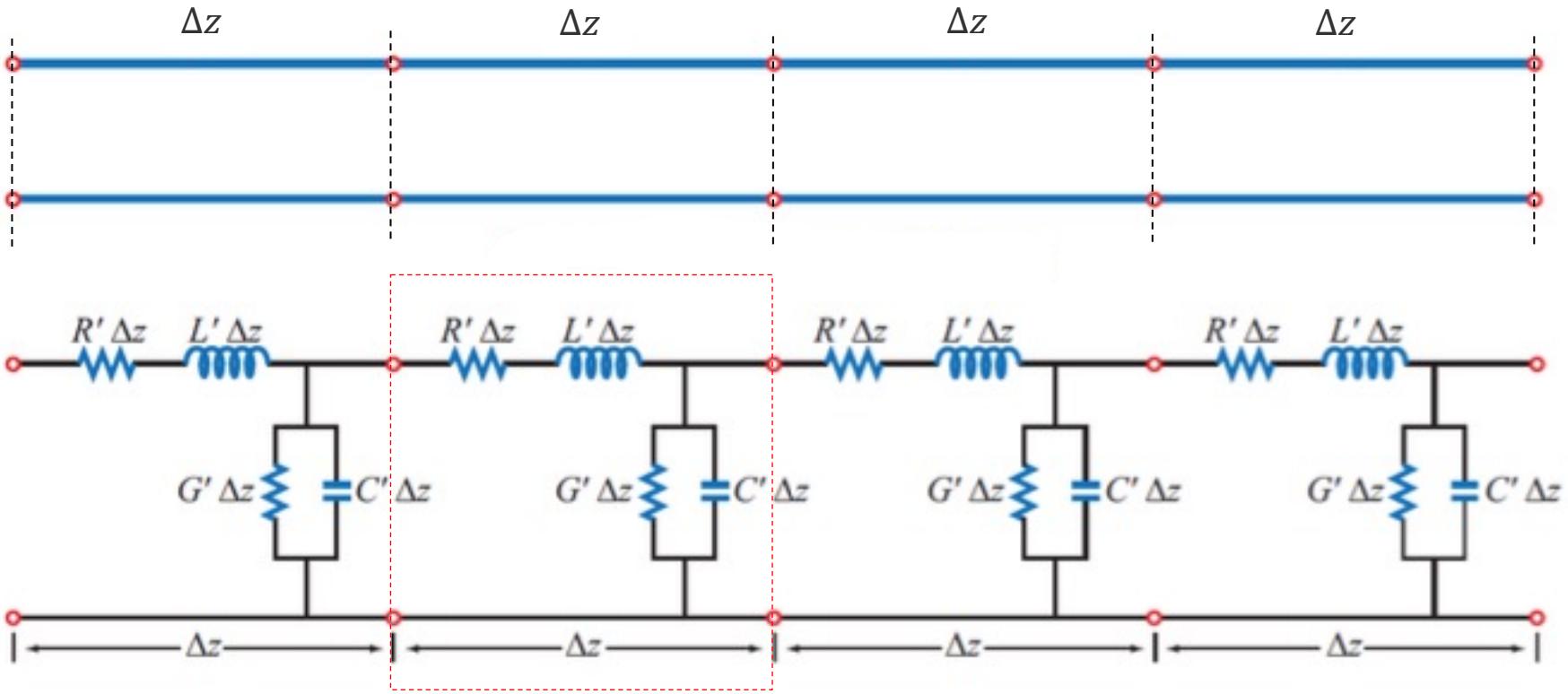
(e) Microstrip line



(f) Coplanar waveguide

TEM Transmission Lines Circuit Model

The circuit model is defined based on "per unit length" elements.



R' Combined resistance of wire Ω/m

G' Combined conductance of insulating material S/m

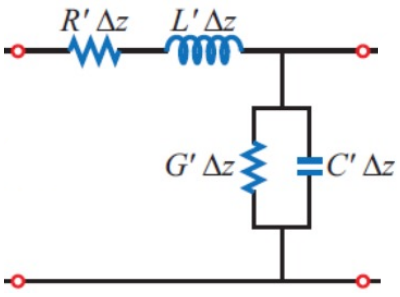
C' Combined capacitance of structure F/m

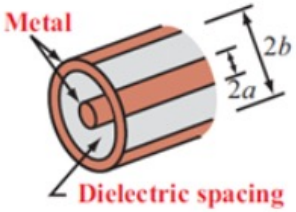
L' Combined inductance of structure H/m

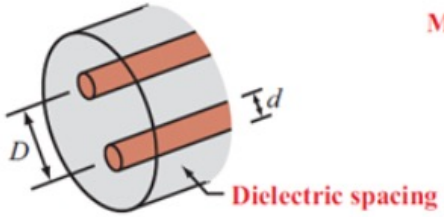
TEM Transmission Lines Circuit Model

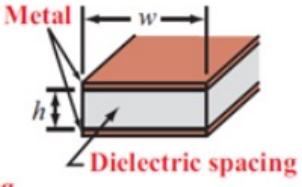
The "per unit length" elements are obtained from field analysis.

Parameter	Coaxial	Two-Wire	Parallel-Plate	Unit
R'	$\frac{R_s}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L'	$\frac{\mu}{2\pi}\ln(b/a)$	$\frac{\mu}{\pi}\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]$	$\frac{\mu h}{w}$	H/m
G'	$\frac{2\pi\sigma}{\ln(b/a)}$	$\frac{\pi\sigma}{\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]}$	$\frac{\sigma w}{h}$	S/m
C'	$\frac{2\pi\varepsilon}{\ln(b/a)}$	$\frac{\pi\varepsilon}{\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]}$	$\frac{\varepsilon w}{h}$	F/m





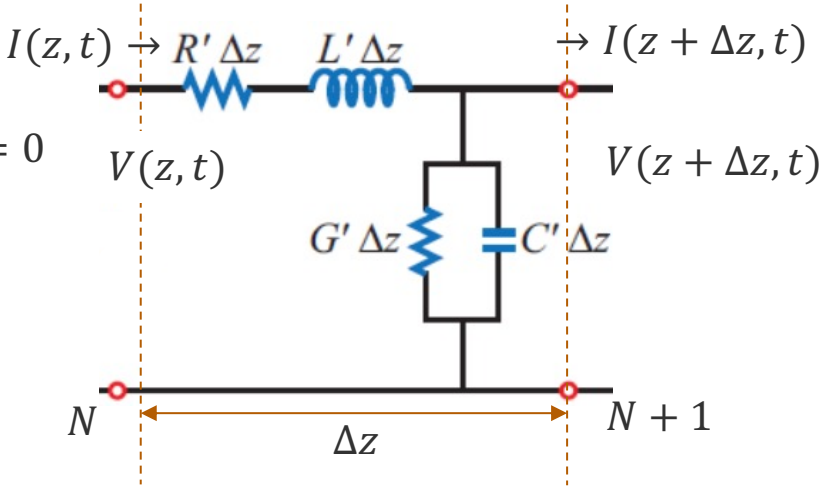




For TEM Lines $L'C' = \varepsilon\mu$ $G'/C' = \sigma/\varepsilon$

Transmission Line Equations

For the unit cell, we can write KVL and KCL.



$$V(z,t) - R'\Delta z\, I(z,t) - L'\Delta z\, \frac{dI(z,t)}{dt} - V(z+\Delta z,t) = 0$$

$$\frac{V(z+\Delta z,t) - V(z,t)}{\Delta z} + R'\, I(z,t) + L'\, \frac{dI(z,t)}{dt} = 0$$

$$-\frac{dV(z,t)}{dz} = R'\, I(z,t) + L'\, \frac{dI(z,t)}{dt}$$

$$I(z,t) - G'\Delta z\, V(z+\Delta z,t) - C'\Delta z\, \frac{dV(z+\Delta z,t)}{dt} = I(z+\Delta z,t)$$

$$-\frac{dI(z,t)}{dz} = G'\, V(z+\Delta z,t) + C'\, \frac{dV(z+\Delta z,t)}{dt}$$

For time harmonic sources, $V(z,t) = V(z)e^{j\omega t}$

$I(z,t) = I(z)e^{j\omega t}$

$$\frac{dV(z)}{dz} = -(R' + j\omega L')\, I(z)$$

$$\frac{dI(z)}{dz} = -(G' + j\omega C')\, V(z)$$

Wave Propagation on a Transmission Line

Insert the two equations into each other.

$$\frac{d^2V(z)}{dz^2} - \gamma^2 V(z) = 0$$

$$\frac{d^2I(z)}{dz^2} - \gamma^2 I(z) = 0$$

$$\gamma^2 = (R' + j\omega L')(G' + j\omega C')$$

$$\gamma = \alpha + j\beta \qquad \alpha = Re\{\gamma\} \qquad \beta = Im\{\gamma\}$$

The solutions are of the form,

$$V(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}$$

$$I(z) = I_o^+ e^{-\gamma z} + I_o^- e^{\gamma z}$$

$$I(z) = \frac{1}{Z_o}(V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z})$$

Also, from the line equations, we write

$$I(z) = \frac{\gamma}{R' + j\omega L'}(V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z})$$

$$I_o^+ = \frac{\gamma V_o^+}{R' + j\omega L'} \qquad I_o^- = \frac{-\gamma V_o^-}{R' + j\omega L'}$$

Or,

$$I_o^+ = \frac{(G' + j\omega C')V_o^+}{\gamma} \qquad I_o^- = \frac{-(G' + j\omega C')V_o^-}{\gamma}$$

We can define "line impedance" as,

$$Z_o = \frac{V_o^+}{I_o^+} = -\frac{V_o^-}{I_o^-}$$

$$Z_o = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

Characteristic Impedance (Ω)

$$Z_o = R_o + jX_o$$

Transmission Line Examples

Lossless Line

$$G' = 0$$

$$R' = 0$$

$$\gamma = \sqrt{(j\omega L')(j\omega C')}$$

$$\alpha = 0$$

$$\beta = \omega\sqrt{L'C'} = \omega\sqrt{\mu\varepsilon}$$

$$Z_o = \sqrt{\frac{L'}{C'}}$$

$$v = c/\sqrt{\varepsilon_r}$$

$$\lambda = 2\pi/\beta$$

Distortionless Line

Attenuation is frequency-independent
Phase is linearly dependent of frequency

$$\frac{R'}{L'} = \frac{G'}{C'}$$

$$\gamma = \sqrt{R'G'}\left(1 + \frac{j\omega C'}{G'}\right)$$

$$\alpha = \sqrt{R'G'}$$

$$\beta = \omega\sqrt{L'C'} = \omega\sqrt{\mu\varepsilon}$$

$$Z_o = \sqrt{\frac{R'}{G'}} = \sqrt{\frac{L'}{C'}}$$

Examples

1) Design a 50-Ohm air line at 700 MHz with phase constant of 20 rad/m.

$$\frac{\beta}{Z_o} = \omega C' = \frac{20}{50} \quad \Rightarrow \quad C' = \frac{20}{50\,\omega} = 90.9 \;\; pF/m$$

$$L' = {Z_o}^2\, C' = 227 \;\; nH/m$$

2) Find the circuit elements of a 60-Ohm distortionless line with attenuation constant of 20 mNp/m and wave speed of $0.6c$.

$$\alpha Z_o = \sqrt{R'G'}\sqrt{\frac{R'}{G'}} = R' \quad \Rightarrow \quad R' = 1.2 \;\; \Omega/m$$

$$\frac{Z_o}{v} = \frac{\sqrt{\frac{L'}{C'}}}{1/\sqrt{L'C'}} = L' \quad \Rightarrow \quad L' = 333 \;\; nH/m$$

$$\frac{\alpha}{Z_o} = \frac{\sqrt{R'G'}}{\sqrt{\frac{R'}{G'}}} = G' \quad \Rightarrow \quad G' = 0.33 \;\; mS/m$$

$$C' = 92.6 \;\; pF/m$$

Q & A

Send me an email if you have any questions.